A Systematic Evaluation of Prompt Design for LLM-Based Sexism Detection

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Abstract. Prompt design offers an appealing route to sexism detection with large language models: no labeled data, no fine-tuning, and a wide space of role, context, and reasoning instructions to choose from. This paper tests that route systematically, evaluating a curated grid of 18 prompt variants across six open-weight, instruction-tuned models spanning five model families and a 3.8–9B parameter range, on the full official EDOS sexism-detection test split ($n = 4{,}000$). Prompt design produces large, statistically real effects: 92 of 101 variant-versus-baseline comparisons reach significance under McNemar's test and 88 of 101 under a paired F1-diff bootstrap, and the single largest of these tested effects comes from prompt design alone. But that entire space sits, for five of the six models, below what a classical TF-IDF and logistic regression classifier already achieves on the same data at a fraction of the computational cost; only one model edges past it on F1, at 4–16× the inference cost of every other prompted model, and a separate confidence-based AUC comparison crowns a different model instead, not the F1 leader. Reading raw model output rather than trusting aggregate scores further revealed that one model's chain-of-thought failures split cleanly into two distinct causes, explicit safety refusal or reasoning-token-budget exhaustion, depending on prompt composition, and that prompt-component order matters far less than prior work on few-shot demonstration order would predict. Prompt engineering is a real, measurable lever on this task; it is not, on this evidence, a substitute for adapting a model to the task's own labeled data.

Keywords: sexism detection · prompt engineering · large language models · in-context learning · chain-of-thought prompting · hate speech detection

1 Introduction

Online sexism is widespread, and automated detection is one of the tools platforms and researchers use to make identifying it tractable at scale [7]. The Explainable Detection of Online Sexism (EDOS) benchmark, evaluated throughout this paper, was built to move that detection effort past a coarse sexist/not-sexist label toward a fine-grained taxonomy of how sexism shows up in text. Most computational work on EDOS, and on hate speech detection more broadly, still relies on fine-tuning a model on labeled examples. Instruction-tuned large language models offer a different route: Brown et al. [3] showed that a sufficiently capable model can perform a new task from a natural-language description and a handful of demonstrations, with no weight update at all. That makes prompting attractive whenever labeled data or compute is scarce.

That option comes with its own open question: which prompt does the job. A prompt for sexism detection can assign the model a role, define the linguistic markers that make language sexist, supply surrounding context about the platform and content type, ask the model to reason before answering, or show worked examples, each a real design choice with no established default for this specific task. Related work on prompt engineering generally, and on hate speech detection with LLMs specifically (Sec. 2), suggests these choices can matter a great deal, but the evidence for sexism detection in particular has so far come from single models evaluated on a narrow slice of that design space, without a non-LLM baseline or formal significance testing to say whether an apparent improvement is real.

This paper closes that gap directly: a curated grid of 18 prompt variants, combining five independently toggleable components (role, aspects, context, chain-of-thought, and few-shot demonstrations) in isolation and in combination, across six open-weight, instruction-tuned models spanning five model families and a 3.8–9B parameter range, on the full official EDOS test split. Every variant's effect against a zero-component baseline is tested for statistical significance with two independent tests, and every model's best training-free variant is compared against a QLoRA-tuned version, a fine-tuned encoder, and a classical TF-IDF and logistic regression baseline, all trained on the same data. Results are further broken down by sexism subtype and tested along three robustness axes prior work has not applied to this task: prompt-component order, few-shot count, and which specific demonstrations are drawn.

The results argue for a narrower and more specific claim than either "prompt design matters" or "LLMs solve sexism detection" taken alone: prompt design produces large, statistically real effects within training-free prompting, but that entire space sits, for five of the six evaluated models, below what a linear classifier trained on the same labeled data already achieves, at a fraction of the computational cost (Sec. 5). Reading raw model output rather than trusting aggregate scores also surfaced a chain-of-thought-triggered failure mode with two distinct causes, and showed that prompt-component order matters far less than prior few-shot-demonstration-order work would predict (Sec. 6). Sec. 7 and Sec. 8 address how far these findings generalize beyond the setting tested here.

2 Related Work

2.1 Sexism and Hate Speech Detection Datasets

The EDOS dataset [7] evaluated throughout this paper was built for SemEval-2023 Task 10, splitting sexism detection into three subtasks of increasing granularity: a binary sexist/not-sexist label, a four-category classification, and an eleven-vector fine-grained taxonomy. Most published EDOS work fine-tunes a pretrained encoder on the training split, the same strategy this paper's own encoder baseline

(Sec. 3.3) follows: shared-task submissions ranged from BERT-based classifiers [12] to conventional machine learning pipelines on hand-engineered features [11], the latter close in spirit to the TF-IDF baseline used here. Where an LLM appears at all in this literature, it is typically one comparison point among several fine-tuned models, not the object of a systematic prompt-design study.

Sexism-specific datasets remain comparatively rare outside English. SWSR [6] is, to our knowledge, the only comparably scaled non-English sexism dataset built with its own dedicated lexicon, drawn from Sina Weibo rather than the Reddit and Gab sources behind EDOS. Its existence is a reminder that the platform and language choices behind EDOS are not incidental: what counts as sexist language, and how directly it gets stated, varies with both.

2.2 Prompt Engineering and In-Context Learning

Brown et al. [3] showed that a sufficiently large frozen language model can pick up a new task from a short natural-language description and a handful of demonstrations, with no gradient update at all. That result opened a design space, mapped by Liu et al.'s survey [8], of choices a practitioner makes when turning a task into a prompt: how to phrase the instruction, whether to supply demonstrations and how many, whether to ask the model to reason before answering. The five toggleable components studied here, role, aspects, context, chain-of-thought, and few-shot (Sec. 3.1), occupy one corner of that space, chosen for a single task rather than surveyed in the abstract.

Chain-of-thought prompting [15] is the best-known instance of the reasoning component in that space, proposed for arithmetic and multi-step logical problems rather than classification. Sexism detection is not obviously the kind of task chain-of-thought was built for, and the results here bear that out unevenly: only two of six models (Sec. 5.2) show a large, consistent benefit from it, which argues against assuming a chain-of-thought instruction helps by default on a new task rather than measuring it directly.

Order matters too. Lu et al. [10] found that permuting the order of few-shot demonstrations alone can move a model from near-state-of-the-art to near-random accuracy on standard classification benchmarks, with no change to which demonstrations are shown. The order sweep in this paper (Sec. 3.5) asks a related but distinct question: not the order of demonstrations within a fixed prompt, but the order of entire prompt components relative to each other (Sec. 6.1 compares the results directly).

2.3 Large Language Models for Hate Speech and Content Moderation

The finding closest in spirit to this paper's own baseline comparison (Sec. 5.5) comes from Plaza-del-Arco et al. [1], who ran several LLMs zero-shot across text classification tasks and found none rivaled simple supervised baselines. This paper reaches a similar conclusion by a different route: an 18-variant grid of systematically combined prompt components, rather than zero-shot alone, so the gap persists after substantial prompt engineering, not only in its absence. A comparable pattern shows up in hate speech detection: Bernal-Beltrán et al. [2] found fine-tuned Spanish encoders consistently ahead of zero- and few-shot LLM prompting, which narrows but does not close the gap. Neither applies this study's statistical testing or multi-model breadth (Sec. 3.4), but both point the same direction: a general-purpose LLM's prompting still lags a model trained on the task's own labels.

Safety alignment introduces a failure mode distinct from accuracy. Röttger et al.'s XSTest [13] is a purpose-built, adversarial suite of safe-sounding prompts designed to trigger unwarranted refusals, and it shows the phenomenon exists in several production models. The llama3.1 refusal behavior documented in this paper (Sec. 5.3) was not sought out through an adversarial test suite: it surfaced as a byproduct of ordinary sexism classification, once role and aspect framing were dropped from the prompt, on genuinely sexist and non-sexist text rather than deliberately borderline phrasing.

2.4 Fine-Tuned Baselines: Encoders and Parameter-Efficient Adaptation

The encoder baseline in this study (Sec. 3.3) fine-tunes RoBERTa [9], whose central finding, that BERT's original training recipe left real performance on the table, shows that a fine-tuning baseline's strength depends on training choices as much as architecture. The QLoRA baseline builds on two more recent ideas: LoRA [5], which adapts a frozen model via small low-rank weight updates instead of retraining all its weights, and QLoRA [4], which quantizes the frozen base weights to 4 bits so the adaptation fits on a single GPU even for a multi-billion-parameter model. Together they make it practical to fine-tune all six of this paper's prompted models rather than only the smallest, which is what makes the QLoRA-versus-prompting comparison in Sec. 5.5 possible at this scale.

3 Methodology

3.1 Prompt-Design Space

We study five independently toggleable prompt components layered on a common base instruction (classify the input text as containing sexism or not): a system-level **role** specification identifying the model as a sexism-annotation assistant; a bulleted list of **aspects** (seven linguistic markers associated with sexist language: objectification, gender stereotyping, hostile language, benevolent sexism, dehumanization, double standards, and dismissiveness); a short **context** block naming the platform, content type, and language; **chain-of-thought** (T) instructions asking the model to reason before answering; and **few-shot** (F) demonstrations.

Rather than the full 2^5=32-cell factorial, we evaluate a curated grid of 18 variants (V1–V18): the bare baseline with every component off (V1), each

Table 1. Evaluated model roster. All Hugging Face Hub revisions are pinned to a fixed commit (Sec. 3.2).

Model	Hugging Face ID	Params	Reasoning
mistral	mistralai/Mistral-7B-Instruct-v0.3	7B	
phi3	microsoft/Phi-3-mini-4k-instruct	3.8B	
llama3.1	meta-llama/Llama-3.1-8B-Instruct	8B	
qwen2.5	Qwen/Qwen2.5-7B-Instruct	7B	
gemma2	google/gemma-2-9b-it	9B	
qwen3	Qwen/Qwen3-8B	8B	✓

component in isolation (V2–V6), and twelve further combinations spanning pairwise through five-way interactions, culminating in all five components active at once (V18). Table 2 (Sec. 5.2) lists the exact component composition of every variant alongside its results, so it is not repeated here.

Active (non-role) components are concatenated into the prompt's user turn in a fixed order (context, chain-of-thought, aspects, few-shot) ahead of the input text; the role component, when active, is instead injected as a separate system-role chat message. Non-chain-of-thought variants append an explicit trailing `ANSWER:` cue immediately before generation, forcing an immediate two-token `YES`/`NO` decision; chain-of-thought variants omit this cue and instead instruct the model to reason first and conclude with its own `ANSWER: YES` / `ANSWER: NO` line, so the forced cue never pre-empts the model's own reasoning (Sec. 4.3 details the corresponding generation-budget difference between the two). Few-shot variants sample k=2 demonstrations per class (2 sexist, 2 not-sexist; 4 total), seeded (seed=42) and drawn without replacement from the EDOS training split (Sec. 4.1), formatted as `TEXT`/`ANSWER` pairs.

3.2 Model Roster

We evaluate six open-weight, instruction-tuned LLMs spanning five model families and a 3.8–9B parameter range (Table 1). Every model is loaded from a Hugging Face Hub revision pinned to a fixed commit hash, so a later upstream update to a repository's default branch cannot silently change which weights a rerun evaluates. Two models (llama3.1, gemma2) are gated, requiring an accepted Hugging Face license; the remaining four are openly accessible. qwen3 is the only reasoning-tuned model in the roster: its chat template exposes a runtime `enable_thinking` toggle, wired to each variant's own chain-of-thought flag, giving it genuine extended reasoning on chain-of-thought variants rather than the brief prose reasoning the other five models produce from instructions alone.

3.3 Fine-Tuned and Classical Baselines

All three baselines below train on the EDOS training split and are scored on the same held-out test split as the prompting grid (Sec. 4.1), for direct comparability with the prompting results (Sec. 5.5).

QLoRA. For each of the six prompted models we additionally fine-tune a QLoRA adapter: 4-bit NF4-quantized frozen base weights (bf16 compute dtype) with LoRA adapters (r=16, α=32, dropout 0.05) targeting the attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`). Training runs 6 epochs (learning rate 2×10^{-4}, batch size 16, max sequence length 512) on the training split, oversampled to a balanced 50/50 class ratio before tokenization: the SFT objective (next-token prediction on a single `" YES"/" NO"` completion) has no built-in class weighting, and, trained directly on EDOS's natural ∼76/24 imbalance, collapses to unconditionally predicting the majority class. Both the fine-tuning prompt template and completion format exactly match variant V1's own zero-component prompt, so the fine-tuned model is compared like-for-like against the prompting-only results at inference time: same instruction, same input format, only the weights differ. We save a per-epoch adapter snapshot and promote the epoch with the highest F1 on the EDOS dev split to the one evaluated on the test split, mirroring the encoder baseline's own selection criterion below.

Encoder. A `roberta-base` sequence classifier (2 output labels), fine-tuned for 3 epochs (learning rate 2×10^{-5}, batch size 64, max sequence length 128) directly on the training split, with the epoch achieving the best dev-split F1 promoted to the reported test-set number.

Classical. A non-neural baseline: TF-IDF features (unigrams and bigrams, minimum document frequency 2, 50,000-feature cap, sublinear term-frequency scaling) feeding a logistic regression classifier (balanced class weighting to offset EDOS's natural imbalance, up to 1,000 solver iterations), fit on the same training split.

3.4 Statistical Testing

For each model we test every non-baseline variant (V2–V18) against V1 with two independent, per-model-family-corrected tests: an exact McNemar's test on the accuracy-based discordant-pair counts between the two variants' predictions, and a paired bootstrap test on the difference in F1 (2,000 resamples, percentile confidence interval). Both tests' p-values are Holm-corrected within their own family of 17 comparisons per model, rather than pooled across the two tests. McNemar and the F1-diff bootstrap test the same baseline-vs-variant hypothesis via different statistics, so correcting one family with the other's p-values would misrepresent the family-wise error control each provides on its own (Sec. 5.4 discusses a case where the two tests disagree).

3.5 Sensitivity-Sweep Design

Three supplementary robustness checks target each model's own best-performing main-grid variant only, not the full grid along a second axis, which would be prohibitively expensive: (i) an **order sweep**, rerendering that variant's active components under up to 5 distinct, seeded random orderings of {context, chain-of-thought, aspects, few-shot} (deduplicated to distinct induced orders, since an inactive component does not affect ordering), holding the active component set itself fixed; (ii) a **few-shot-size sweep**, rerunning the variant at $k \in \{1, 3, 4\}$ demonstrations per class against the grid's own default k=2, for variants where few-shot is active; and (iii) a **demonstration-seed sweep**, redrawing which k=2-per-class demonstrations are sampled under three alternate random seeds (1, 7, 123) against the grid's own fixed seed (42), holding both the component order and the demonstration count fixed, for variants where few-shot is active. All three sweeps reuse the same evaluation engine and generation settings as the main grid (Sec. 4.3); Sec. 5.7 and Sec. 5.8 report the results.

4 Experimental Setup

4.1 Dataset

We use the full official release of the Explainable Detection of Online Sexism (EDOS) dataset from SemEval-2023 Task 10 [7]: 20,000 English-language social media posts labeled sexist / not sexist, with every sexist post further annotated into one of four categories (threats/plans to harm/incitement, derogation, animosity, and prejudiced discussions). We use the dataset's own official train/dev/test partition (14,000/2,000/4,000), fetched from a pinned upstream commit and verified by SHA-256 checksum against the exact release used throughout this study, so an upstream force-push cannot silently change what was evaluated. Class balance is stable at roughly 24% sexist / 76% not sexist across all three splits (train: 3,398 sexist / 10,602 not sexist; dev: 486 / 1,514; test: 970 / 3,030, the exact figures underlying Sec. 5.1's base rate).

The training split supplies both the few-shot demonstration pool (Sec. 3.1) and the fine-tuning data for all three baselines (Sec. 3.3); the dev split is used only for the fine-tuned baselines' epoch selection (Sec. 3.3). All reported metrics (prompting grid, baselines, and sensitivity sweeps alike) are computed on the same held-out test split. For the subtype-level error analysis (Sec. 5.6), the test split's 970 sexist examples break down as: derogation (454), animosity (333), prejudiced discussions (94), and threats/plans to harm/incitement (89).

4.2 Inference Precision

Generation uses bf16 throughout, with no additional post-training quantization: calibration showed 4-bit NF4 quantization and bf16 cost effectively the same wall-clock time per example (30.28s vs. 30.47s), so bf16 was kept as the default given ample VRAM headroom across the roster, rather than paying the complexity of a quantized inference path for no measured throughput benefit. This is separate from the QLoRA baseline (Sec. 3.3), which always quantizes its frozen base weights to 4-bit NF4 as standard QLoRA practice; there, quantization is what makes fine-tuning a full-size 7–9B model on a single GPU feasible at all, an entirely different concern from generation-time speed.

4.3 Generation and Decoding

All generation is greedy (`do_sample=False`, `top_p=1.0`, no repetition penalty) under a fixed global seed (42) throughout. Non-chain-of-thought variants use a 2-token generation budget, matching their forced trailing `ANSWER:` cue (Sec. 3.1); chain-of-thought variants use a 220-token budget, calibrated for the five non-reasoning models' brief prose reasoning. qwen3 is the exception: its native thinking mode can reason for far longer than 220 tokens before reaching a decision, so its chain-of-thought units use a 1,024-token budget instead, directly reflected in its substantially higher per-example latency (Sec. 5.9). Generation batch size is 32 for five of the six models, calibrated to stay within available GPU memory (zero out-of-memory failures observed at that size); qwen3 uses batch size 128, calibrated the same way for its longer generations (fail_ratio 0.008 at that batch size).

4.4 Evaluation Metrics

We report accuracy and binary precision/recall/F1 with the sexist class as positive, computed identically for every (model, variant) unit, baseline, and sweep condition. A response is parsed by locating its *last* occurrence of an answer marker (`ANSWER:`, or a residual chat-template artifact) and reading the YES/NO token that follows; this single rule handles both the prompt-inserted marker of non-chain-of-thought variants and the model-produced marker at the end of a chain-of-thought response without special-casing either. A response that never reaches an unambiguous YES/NO decision (including the deliberately unresolved case of both words appearing with no leading marker) is a parse failure; the fraction of failures is reported per unit as `fail_ratio` and, in the headline metrics used throughout this paper, failures default to a "not sexist" prediction. This default is a real cost only when `fail_ratio` is non-trivial, which is itself informative (Sec. 5.3).

5 Results and Analysis

5.1 Experimental Overview

All results below are computed against the full official EDOS test split [7], $n = 4{,}000$, at its real class distribution: 3,030 not-sexist examples (75.75%) and 970 sexist examples (24.25%). We state this base rate up front because several of the accuracy figures reported in this section are only interpretable against it: a classifier that predicts "not sexist" unconditionally would already score 0.758 accuracy by construction.

5.2 Main Prompt-Design Grid

Table 2 reports F1 for all 18 prompt variants across the six evaluated models. Bold marks the best model for a given variant (row); underline marks each model's own best-performing variant (column).

Table 2. F1 score by prompt variant (V1–V18) and model, full EDOS test split ($n = 4{,}000$). R = role, A = aspects, C = context, T = chain-of-thought, F = few-shot (k=2 per class). Bold: best model per row; underline: best variant per model (column). †llama3.1's V14 omission and the V8/V13/V15 caveat are explained in Sec. 5.3.

	Prompt Features					F1 Score					
Variant	R	A	C	T	F	Mistral-7B	Phi-3-mini	Llama-3.1-8B	Qwen2.5-7B	Gemma-2-9B	Qwen3-8B
V1	–	–	–	–	–	0.467	0.138	0.441	**0.537**	0.448	0.481
V2	✓	–	–	–	–	0.454	0.256	0.472	**0.535**	0.444	0.483
V3	–	✓	–	–	–	0.490	0.413	0.550	**0.570**	0.501	0.543
V4	–	–	✓	–	–	0.453	0.056	0.466	**0.493**	0.454	0.477
V5	–	–	–	✓	–	0.496	0.480	0.412	0.509	0.443	**0.553**
V6	–	–	–	–	✓	0.567	0.517	<u>0.550</u>	<u>0.585</u>	0.530	**0.590**
V7	✓	–	–	–	✓	<u>0.567</u>	0.533	0.519	0.583	<u>0.530</u>	**0.594**
V8	–	–	–	✓	✓	0.471	0.438	0.240	0.571	0.503	<u>**0.627**</u>
V9	✓	✓	–	–	✓	0.559	<u>**0.566**</u>	0.471	0.559	0.514	0.562
V10	✓	✓	–	✓	–	0.461	0.478	0.497	0.543	0.454	**0.544**
V11	–	✓	✓	✓	–	0.483	0.466	0.508	0.556	0.471	**0.559**
V12	–	✓	–	✓	✓	0.462	0.461	0.501	0.568	0.496	**0.602**
V13	✓	–	–	✓	✓	0.473	0.448	0.361	0.576	0.482	**0.614**
V14	–	–	✓	✓	✓	0.447	0.461	†–	0.571	0.500	**0.618**
V15	✓	–	✓	✓	✓	0.452	0.464	0.360	0.581	0.479	**0.618**
V16	✓	✓	✓	✓	–	0.474	0.473	0.495	0.537	0.454	**0.546**
V17	✓	✓	–	✓	✓	0.440	0.486	0.470	0.556	0.485	**0.586**
V18	✓	✓	✓	✓	✓	0.439	0.489	0.494	0.572	0.477	**0.591**

Two patterns hold across all six models. First, the zero-shot baseline (V1) is never any model's best variant, and the fullest five-component prompt (V18) is not either: for every model, the top score comes from a two- or three-component combination that includes few-shot examples. Second, model behavior at V1 varies far more than the eventual ceiling does: qwen2.5 opens at F1 0.537 with no prompt engineering at all, while phi3 opens at F1 0.138, driven by a recall of just 0.082. It flags almost nothing as sexist until given role framing, aspect decomposition, or examples to work from. Qwen2.5 is the strongest model for every V1–V4 (role/aspects/context-only) variant, but qwen3 becomes the strongest model for every variant from V5 onward, once chain-of-thought or few-shot enters the prompt. Qwen3's own best variant, V8 (chain-of-thought plus few-shot, F1 0.627), is also the single highest score in the entire grid.

5.3 Qualitative Finding: Safety-Alignment Interference in Llama-3.1-8B

Llama-3.1-8B's V14 (context, chain-of-thought, few-shot; no role or aspect framing) is omitted from Table 2 rather than scored: 54.6% of its 4,000 generations do not reach a parseable YES/NO decision, reproduced identically across two independent runs under this study's deterministic (greedy) decoding. Reading the raw generations, these are not truncated or malformed outputs: they are explicit safety refusals (e.g., "I cannot create content that is discriminatory or hateful"). Chain-of-thought is the trigger: no variant without it shows a meaningful `fail_ratio` for this model. V14 is the worst single case of a pattern that

Table 3. `fail_ratio` for llama3.1's ten other chain-of-thought variants. Refusal share is the fraction of a variant's failed rows matching an explicit decline pattern (e.g. "I cannot," "not able to"); the remainder are token-budget truncations, not declines (see text).

Variant	Components	Aspects?	fail_ratio	Refusal share
V8	T+F	–	0.491	0.98
V15	R+C+T+F	–	0.441	0.94
V13	R+T+F	–	0.395	0.90
V5	T	–	0.236	0.82
V11	A+C+T	✓	0.214	0.31
V12	A+T+F	✓	0.214	0.54
V17	R+A+T+F	✓	0.171	0.63
V16	R+A+C+T	✓	0.166	0.46
V18	R+A+C+T+F	✓	0.146	0.32
V10	R+A+T	✓	0.121	0.45

runs across every chain-of-thought variant llama3.1 was evaluated on, detailed below, not an isolated one.

Scoring refusals as "not sexist," the standard convention this study uses elsewhere, V14 reaches 0.726 accuracy (almost exactly the test split's 0.758 not-sexist base rate), with recall collapsing to 0.278 and F1 at 0.330. Restricted to the 45.4% of rows the model actually answers, F1 recovers to 0.525, in line with llama3.1's better-scoring variants (V3 and V6, both F1 0.550). The two views tell different stories: the model does not appear worse at the task when it engages, but it declines to engage on over half the inputs under this specific prompt structure. The answered subset is not a random sample either: refusal plausibly correlates with how explicit the source text is. This is a finding about llama3.1's safety alignment interacting with prompt structure, not a gap in the evaluation; full derivation and the two-run reproduction are documented alongside the raw refusal transcripts in the project's supplementary material.

Table 3 reports `fail_ratio` for llama3.1's ten other chain-of-thought variants. Classifying each variant's failed rows by whether they match an explicit refusal pattern reveals two distinct causes, not one, and which dominates splits cleanly on whether aspects is active. Every variant without aspects is refusal-dominant (82–98% of its failures are explicit declines, averaging around 100 characters each, matching V14's own transcripts). Every variant with aspects is truncation-dominant or close to an even split (31–63% refusal); its non-refusal failures average 1,000–1,080 characters, sitting right at this study's 220-token chain-of-thought budget (Sec. 4.3), and on inspection consist of genuine step-by-step reasoning cut off mid-sentence before reaching a decision, not a truncated refusal message. Aspects asks the model to work through seven named linguistic markers before answering; for llama3.1, that reasoning routinely does not finish inside a token budget calibrated for the other five models' shorter prose reasoning. V8 (`fail_ratio` 0.491), V13 (0.395), and V15 (0.441) are the three most refusal-

Table 4. Significance of prompt-design effects vs. baseline V1, Holm-corrected within each model ($\alpha = 0.05$). "Compared" excludes V1 itself and, for llama3.1, the omitted V14. "McN. sig."/"F1 sig." are the counts of variants significant under exact McNemar's test and the paired F1-diff bootstrap, respectively (Sec. 3.4).

Model	Compared	McN. sig.	F1 sig.	Best var.	Best ΔF1
Mistral-7B	17	17	12	V7	+0.100
Phi-3-mini	17	13	17	V9	+0.428
Llama-3.1-8B	16	16	16	V6	+0.110
Qwen2.5-7B	17	14	14	V6	+0.049
Gemma-2-9B	17	16	14	V7	+0.082
Qwen3-8B	17	16	15	V8	+0.146
Total	101	92	88	–	–

affected cells still reported in Table 2 (F1 0.240, 0.361, and 0.360 respectively); their scores carry the same caveat as V14's, just below rather than above the exclusion threshold: a property of this specific model, worth knowing before comparing its numbers to the other five.

5.4 Statistical Significance of Prompt-Design Effects

For each model we test every non-baseline variant against V1 with exact McNemar's test (on the discordant-pair counts between the two variants' predictions) and a paired bootstrap test on the F1 difference (2,000 resamples), both Holm-corrected across the model's variants. Table 4 summarizes the results.

88 of the 101 variant-vs-baseline comparisons across all six models show a significant F1 difference, and 92 of 101 show a significant McNemar effect: the large majority of this study's prompt-design choices produce a real, statistically distinguishable effect on model behavior, not sampling noise. The two tests disagree on a substantial fraction of comparisons, in both directions (mistral: 17 McNemar-significant vs. 12 F1-diff-significant; phi3: 13 vs. 17). This divergence is expected, since the two tests measure different quantities; phi3's V1→V9 jump, below, shows why.

Phi-3-mini's V1→V9 jump is the clearest illustration. F1 rises from 0.138 to 0.566 (+0.428, by far the largest single effect in this study), while accuracy moves only from 0.750 to 0.759 and the McNemar test does not reach significance after correction ($p_{\text{holm}} = 0.695$). The reason is arithmetic, not a contradiction between tests: at V1, recall is 0.082 and precision 0.421, implying on the order of 80 true positives against 110 false positives out of 4,000 predictions. At V9, recall reaches 0.647 and precision 0.502, implying roughly 628 true positives against 622 false positives. The variant gains around 548 correct sexist-flags and loses around 512 new false alarms, almost, but not exactly, offsetting on a 4,000-row test set that is 76% negative, so raw accuracy (and the McNemar test built on it) barely moves. F1, which is computed against the 970-row positive class rather than the full test set, is not diluted by that majority-class denominator, so it

Table 5. Best training-free prompting vs. fine-tuned and classical baselines. Classical (TF-IDF + logistic regression): F1 0.617, accuracy 0.799. Encoder (RoBERTa-base): F1 0.735, accuracy 0.874. Δ columns are F1 relative to the classical baseline.

Model	Best prompt F1	QLoRA F1	Δ vs. classical
Mistral-7B	0.567 (V7)	0.788	−0.050
Phi-3-mini	0.566 (V9)	0.734	−0.051
Llama-3.1-8B	0.550 (V6)	0.795	−0.067
Qwen2.5-7B	0.585 (V6)	0.786	−0.032
Gemma-2-9B	0.530 (V7)	0.794	−0.087
Qwen3-8B	0.627 (V8)	0.779	+0.010

reflects the same underlying shift as a large, clearly significant change. This is the concrete case for reporting both test families rather than collapsing them into one number, and it is the opposite pattern from every other model in this study, where McNemar catches more significant comparisons than the F1-diff bootstrap does.

5.5 Comparison Against Fine-Tuned and Classical Baselines

Table 5 places each model's best training-free prompt variant against a QLoRA-tuned version of the same model, a single TF-IDF plus logistic-regression classifier (no LLM), and a fine-tuned RoBERTa-base encoder.

The headline result of this comparison is not favorable to prompting. Five of the six models' best training-free prompt variant, after an 18-point search over prompt design, scores below a bag-of-words logistic regression classifier: a baseline with no language model, no GPU, and a fraction of a second of CPU inference per example. Only qwen3 edges past it, by F1 +0.010, and even qwen3 loses to it on accuracy (0.773 vs. 0.799). QLoRA fine-tuning and the RoBERTa encoder are in a different tier entirely: every QLoRA-tuned model reaches F1 0.734–0.795, and the encoder reaches F1 0.735, both comfortably above the classical baseline and far above every prompted number. Ranked by F1, the ordering is consistent across the whole study: QLoRA $\approx$ encoder $\gg$ classical $>$ training-free prompting, with qwen3 as the one partial exception at the classical boundary. This is the direct answer to this study's "no non-LLM baseline" gap: the classical comparison reframes what the prompt-design results in Sec. 5.2–5.4 mean. Prompt engineering produces large, statistically real differences *within* the space of training-free approaches, but that entire space sits, for five of six models, below what a linear classifier over word counts already achieves on this task.

5.6 Error Analysis by Sexism Subtype

Table 6 breaks down each model's best variant by the four fine-grained sexism categories in the EDOS taxonomy [7], reporting the false-negative rate (share of truly sexist examples in that category the model misses) for each.

Table 6. False-negative rate by sexism subtype, best variant per model. 1: threats/plans to harm/incitement ($n = 89$). 2: derogation ($n = 454$). 3: animosity ($n = 333$). 4: prejudiced discussions ($n = 94$).

Model	Variant	1	2	3	4	Mean
Mistral-7B	V7	0.011	0.057	0.147	0.181	0.099
Phi-3-mini	V9	0.337	0.300	0.372	0.553	0.391
Llama-3.1-8B	V6	0.079	0.064	0.129	0.255	0.132
Qwen2.5-7B	V6	0.281	0.141	0.303	0.309	0.258
Gemma-2-9B	V7	0.022	0.015	0.048	0.043	0.032
Qwen3-8B	V8	0.303	0.150	0.237	0.372	0.266

Table 7. Sensitivity of each model's best variant to component reordering and few-shot count, as ΔF1 from the Table 2 value. n/a: variant not eligible for order sweep (see text).

Model (variant)	Order	k=1	k=3	k=4
Mistral-7B (V7)	n/a	−0.039	+0.024	+0.028
Phi-3-mini (V9)	−0.032	+0.024	−0.019	−0.092
Llama-3.1-8B (V6)	n/a	+0.023	−0.004	+0.001
Qwen2.5-7B (V6)	n/a	−0.031	−0.005	−0.002
Gemma-2-9B (V7)	n/a	−0.018	−0.011	+0.011
Qwen3-8B (V8)	+0.004	−0.005	−0.008	+0.018

Even at each model's own best variant, subtype-level performance diverges far more than the aggregate F1 numbers in Table 2 suggest. Gemma-2-9B misses under 5% of threats and derogation cases; phi3, at its own best variant, still misses 55% of "prejudiced discussions" and over a third of every other category. Category 4 (prejudiced discussions, the smallest class at $n = 94$) is the hardest subtype for five of the six models. This gap widens sharply when the worst rather than the best variant is considered: phi3's worst variant (V4, context-only) misses 96–100% of sexist examples in every subtype, a near-total collapse into predicting "not sexist," while gemma2's worst variant (V8) never exceeds a 16% false-negative rate in any subtype. The spread between a model's best-case and worst-case prompt is itself a per-model property, not a constant: gemma2 is the most robust to a bad prompt choice in this study, and phi3 the least.

5.7 Sensitivity to Prompt-Component Order and Few-Shot Size

For each model's best main-grid variant, we re-evaluate under (a) an alternate ordering of its prompt components, where the variant has more than one component the harness can meaningfully reorder, and (b) an alternate few-shot count ($k \in \{1, 3, 4\}$ per class, against the main grid's default $k = 2$). Table 7 reports each as an F1 delta from the main-grid value.

Table 8. Sensitivity of each model's best variant to which few-shot demonstrations are drawn: F1 range across 3 alternate seeds (1, 7, 123), as ΔF1 from the Table 2 value (grid's own seed, 42).

Model (variant)	Grid F1	ΔF1 range
Mistral-7B (V7)	0.567	−0.025 to +0.011
Phi-3-mini (V9)	0.566	−0.235 to +0.007
Llama-3.1-8B (V6)	0.550	−0.097 to −0.028
Qwen2.5-7B (V6)	0.585	−0.037 to −0.011
Gemma-2-9B (V7)	0.530	−0.034 to −0.012
Qwen3-8B (V8)	0.627	−0.025 to −0.003

Order sensitivity applies only to phi3 (V9: role, aspects, few-shot) and qwen3 (V8: chain-of-thought, few-shot). The other four models' best variants either combine a single component with few-shot or combine two components that render identically regardless of order under this study's prompt template, so no meaningful reordering exists to test. Where it does apply, the two models diverge sharply: qwen3 is stable under reordering ($\Delta = +0.004$), while phi3 loses 0.032 F1 under the alternate order, a real, if modest, position effect for this model specifically.

The few-shot-size sweep carries a more general caveat. This study's main-grid results all use $k = 2$ examples per class, but that is not a tuned optimum: four of six models score higher F1 at $k = 4$ than at the $k = 2$ default (mistral +0.028, gemma2 +0.011, qwen3 +0.018, llama3.1 +0.001), meaning the headline numbers in Table 2 understate what few-shot prompting can achieve for most of this study's models. Phi-3-mini is the exception, and a stark one: F1 falls monotonically as k increases past the default, down to −0.092 at $k = 4$, driven by precision rising (0.530→0.670) as recall collapses (0.663→0.366). Where the other five models either improve or stay roughly flat with more examples, phi3 becomes systematically more conservative: the same under-triggering behavior visible in its V1 baseline (Sec. 5.2) re-emerges as few-shot count grows, rather than being fixed by it.

5.8 Sensitivity to Demonstration Selection

The order and size sweeps above vary how the few-shot demonstrations are arranged and how many are shown, but never which specific examples are drawn: every main-grid variant samples its k=2-per-class demonstrations once, with a fixed seed (42). Table 8 reruns each model's best variant under three alternate seeds (1, 7, 123), reporting the resulting F1 range as a delta from the main-grid value.

Four of six models stay within a narrow, consistently negative range (gemma2, qwen2.5, and qwen3 within 0.02–0.04 F1; llama3.1 within 0.03–0.10 F1): every alternate seed tested scores at or below the main grid's own draw. Phi-3-mini is again this study's outlier, and by the widest margin observed in any sensitivity

Table 9. Inference cost across the full 18-variant grid, per model.

Model	Total (h)	Mean (s/ex.)	Slowest variant
Mistral-7B	3.92	0.196	V10, 25.9 min
Phi-3-mini	3.19	0.160	V12, 24.0 min
Llama-3.1-8B	4.71	0.249	V18, 31.5 min
Qwen2.5-7B	1.51	0.075	V11, 14.0 min
Gemma-2-9B	6.26	0.313	V18, 38.5 min
Qwen3-8B	23.92	1.196	V18, 146.8 min

check reported here: F1 falls from 0.566 to 0.330 under one alternate seed. This is not a parsing failure, `fail_ratio` is 0 on every seed tested, main grid included, but a genuine behavioral shift: precision rises to 0.744 while recall collapses to 0.212, the same conservative, under-triggering pattern already observed as k grows past the default (Sec. 5.7) and in this model's V1 baseline (Sec. 5.2), now shown to also depend on which four examples happen to be drawn. The main grid's own seed scores at or above all three alternate seeds for four of six models (llama3.1, qwen2.5, gemma2, qwen3); the two exceptions are mistral, where seeds 1 and 7 each score marginally higher (+0.011, +0.002 F1), and phi3, where seed 7 scores 0.007 F1 higher despite this study's largest sensitivity-sweep swing in the opposite direction coming from a different seed on the same model. Seed 42 was fixed as an arbitrary default before this sweep was run, not chosen for its result, but the pattern means Table 2's headline numbers sit, for most models, closer to the top than the middle of what a different demonstration draw would produce.

5.9 Computational Cost

Table 9 reports wall-clock inference cost for the full 18-variant grid per model, recorded automatically as part of every run.

Qwen3-8B is an outlier by a wide margin: its full grid takes roughly 5–16× longer than any other model's, and its mean per-example cost (1.196s) is 4–16× every other model's. This is a direct consequence of its native reasoning mode, which this study leaves enabled for chain-of-thought variants rather than capping its token budget to match the other five models. Qwen3 is also this study's best-performing model overall (Table 2, 4); that performance is not free, and the accuracy/cost trade-off it represents is a separate question from whether it is the strongest model on accuracy alone.

5.10 Confidence-Based AUC: A Second Comparison Against the Classical Baseline

For the seven non-chain-of-thought variants (V1–V4, V6, V7, V9), the model's decision is its first generated token, so a calibrated P(sexist) score can be read directly off that token's YES/NO logits, softmax-normalized between the two

Table 10. AUC from first-token P(sexist), best non-chain-of-thought variant per model. Classical baseline (TF-IDF + logistic regression, `predict_proba`): AUC 0.848. Δ is AUC relative to the classical baseline. †Qwen3-8B's true best variant, V8 (F1 0.627, Table 5), is chain-of-thought and has no single decision token this method can score; V7 is its best scorable stand-in, not its true best variant.

Model	Variant	F1	AUC	Δ vs. classical
Mistral-7B	V7	0.567	0.843	−0.005
Phi-3-mini	V9	0.566	0.810	−0.038
Llama-3.1-8B	V6	0.550	0.832	−0.016
Qwen2.5-7B	V6	0.585	0.812	−0.036
Gemma-2-9B	V7	0.530	0.875	+0.027
Qwen3-8B†	V7	0.594	0.838	−0.010

candidates, without generating anything further. Chain-of-thought variants are excluded: their decision only emerges after free-form reasoning, so no single generated token stands in for it. We extract this score for all six models across the full 4,000-example test split and score it against the true labels with ROC-AUC. As a validity check, the confidence-implied call (P(sexist) $\geq$ 0.5) agrees with the model's actual generated decision on 94.5–99.7% of rows for every model's variant in Table 10, confirming the first-token assumption holds on real output, not only by construction.

Table 10 complicates the story in Table 5 rather than repeating it. On F1, qwen3 is the only model to beat the classical baseline, and it does so with V8, a chain-of-thought variant this AUC method cannot score at all. On AUC, qwen3's best scorable variant loses to the classical baseline (0.838 vs. 0.848), while gemma2, whose F1 (0.530) trails the classical baseline by the widest margin in this study, is the only model whose AUC (0.875) beats it. Which model, if any, "beats" the classical baseline depends on which metric answers the question. Training-free prompting still fails to consistently clear the classical baseline under either metric; the exception to that just changes identity depending on which metric is used.

5.11 Confidence-Based Abstention: Trading Coverage for Precision

The same first-token P(sexist) score behind Table 10 also supports a simple abstention policy: instead of forcing a decision on every example, withhold a prediction whenever the score falls within a symmetric band around the 0.5 decision boundary, deferring the closest calls rather than answering them. For each model's Table 10 variant, Table 11 reports the resulting coverage (the fraction of the test split still answered) and F1 on that covered subset at a $\pm$0.2 band, against the same variant's full-coverage F1 (band=0) for reference.

F1 improves for every model at this band. Beyond being calibrated in aggregate (Table 10), the confidence score tracks per-example difficulty: the examples it flags as uncertain really are, on average, more error-prone. The size of that

Table 11. Confidence-based abstention at a ± 0.2 band around P(sexist)=0.5, same variant per model as Table 10. Coverage is the fraction of the 4,000-example test split still answered after abstention; F1 is computed on that covered subset only; ΔF1 is relative to the same variant's full-coverage (no-abstention) F1.

Model	Variant	Coverage	F1	ΔF1
Mistral-7B	V7	0.941	0.585	+0.015
Phi-3-mini	V9	0.628	0.691	+0.109
Llama-3.1-8B	V6	0.809	0.601	+0.049
Qwen2.5-7B	V6	0.971	0.593	+0.007
Gemma-2-9B	V7	0.959	0.541	+0.011
Qwen3-8B	V7	0.962	0.602	+0.008

improvement, and the coverage it costs to get it, is not uniform. Phi-3-mini gains the most (+0.109 F1) but only by abstaining on 37% of the test split (coverage 0.628), a policy this study's other five models never need to reach a comparable gain: mistral, gemma2, qwen2.5, and qwen3 all retain over 94% coverage at the same band while gaining under 0.02 F1. Llama-3.1-8B falls between the two patterns (+0.049 F1 at 81% coverage). This mirrors a pattern already visible elsewhere in this study: phi3's errors are not spread evenly across the test set but concentrated in a recoverable subset (Sec. 5.7, Sec. 5.8, and Sec. 5.6 all single out phi3 as an outlier among the six models), and abstention is one more lens that finds the same thing. As with Table 10, this policy is evaluated only on the seven non-chain-of-thought variants; whether the same abstention gain holds for chain-of-thought variants, where no single token yields a confidence score, is untested by this method.

6 Discussion

6.1 What "Prompt Design Matters" Actually Means Here

The statistical results (Sec. 5.4) show a real, measurable shift in model behavior for the large majority of this study's prompt-design choices: phi3's role-plus-aspects-plus-few-shot jump (+0.428 F1, V1→V9) is the largest single effect this study's significance tests measure. Fine-tuning was never put through the same tests, but for this same model the QLoRA gain is larger still: F1 0.734 against V1's 0.138 (Table 5), a +0.596 improvement. Prompt design's effect is large by the standards of training-free methods; whether it is also large against adapting the model's own weights is answered by the baseline comparison (Sec. 5.5): those effects operate entirely within a space that, for five of six models, tops out below a linear classifier trained on the same 14,000 examples in a fraction of a second on a CPU. Prompt design is a real lever on this task, but it is attached to a ceiling a much simpler method already clears.

That ceiling moves, too: four of six models score higher at $k = 4$ than the main grid's own $k = 2$ default (Sec. 5.7), so Table 2 understates each model's real ceiling. Recomputing the baseline gap at each model's best available k does not close it (mistral $-0.050 \rightarrow -0.022$; gemma2 $-0.087 \rightarrow -0.076$; llama3.1 $-0.067 \rightarrow -0.066$, all still below classical; qwen3, already ahead, extends its lead to $+0.028$): the honest-limits finding survives its own most favorable adjustment, only its margin moves.

One axis mattered less than prior in-context-learning work on other tasks would predict. Lu et al. [10] found that reordering few-shot demonstrations alone can swing a model between near-state-of-the-art and near-random performance. Reordering entire prompt components produced nothing like that here: four of six models had no meaningfully reorderable best variant at all, and where reordering did apply, only phi3 showed a real effect (-0.032 F1). Component-order stability is a genuine, if narrow, piece of good news for anyone assembling a prompt in this space: which components to include is what matters, not primarily what order to put them in.

6.2 The Cost of the One Exception

qwen3 is the sole model whose best training-free prompt beats the classical baseline on F1, and the paper's own framing needs to be honest about what that win costs. Table 9 shows qwen3's full grid running 5–16× longer than any other model's, driven by its native reasoning mode left unconstrained on chain-of-thought variants (Sec. 4.3), for a margin of only $+0.010$ F1 over a classical baseline that runs in a fraction of a second per example with no GPU at all, and qwen3 still trails that baseline on accuracy (0.773 vs. 0.799). Even the one model that wins does not win by much, or cheaply.

That framing holds under F1 specifically; it does not hold under every lens. Sec. 5.10's confidence-based AUC comparison shows qwen3's best scorable variant losing to the classical baseline (0.838 vs. 0.848), while gemma2, whose F1 trails classical by the widest margin in this study, is the one model whose AUC beats it. "The one exception" depends on which metric answers the question, and qwen3's true best variant (V8) cannot be evaluated by the AUC method at all: one honest answer among two, not a weaker version of the cost argument above.

6.3 Safety Alignment as an Operational Risk

Llama-3.1-8B's V14 behavior (Sec. 5.3) is easy to misread as a parsing bug or a prompt-engineering mistake specific to one variant. It is neither: the model reads a sexism-classification request, once asked to reason step by step before answering, as a request to generate hateful content rather than analyze it, and declines on safety grounds on over half of V14's rows, at lower but still substantial rates across every other chain-of-thought variant it was evaluated on. Scored under this study's convention, that decline registers as an unremarkable F1 of 0.330, indistinguishable from an ordinary weak variant until the raw generations, not just the aggregate metric, were inspected. Röttger et al.'s XSTest [13] showed the exaggerated-safety version of this failure mode under adversarial probing; llama3.1's refusals show it can also emerge unprompted, from an ordinary classification pipeline, on genuinely on-topic content. For any deployment that scores a refusal as a default label, this is a silent failure mode: no error, just a plausible-looking, systematically wrong answer on a substantial share of one input class. `fail_ratio` (Sec. 4.4), not accuracy or F1 alone, is what surfaces it, and only because this study checked.

6.4 What a Single Aggregate Score Hides

The subtype breakdown and the significance-testing disagreement point at the same underlying problem. The subtype breakdown (Sec. 5.6) shows a model's own best variant can still miss over half of the rarest sexism category, prejudiced discussions, while its aggregate F1 looks unremarkable: gemma2 and phi3 post worst-variant false-negative spreads of 16% and 96–100% respectively, a gap the main grid's F1 column alone never reveals. The significance-testing disagreement (Sec. 5.4) reflects the same class-imbalance sensitivity: phi3's largest F1 effect in the entire study is invisible to accuracy-based McNemar testing, because a 76%-negative test set can absorb a large shift in positive-class detection without moving overall accuracy much at all. Röttger et al.'s HateCheck [14] made the general version of this argument for hate speech detection: held-out accuracy and F1 can look strong while hiding specific, systematic weaknesses that only a finer-grained test reveals. This study's subtype table and its dual significance tests apply that same argument to sexism detection, in two independent places rather than one.

7 Limitations

This study covers a single dataset, language, and pair of source platforms: EDOS's English-language Reddit and Gab posts [7]. Sexist language is expressed differently across languages and platforms (Sec. 2.1 discusses SWSR [6]), so nothing here establishes that the same prompt-design effects, the same baseline gap, or the same safety-refusal pattern would reproduce on a different dataset. The study also targets only EDOS's binary sexist/not-sexist subtask directly; the finer four-category taxonomy is used solely for the post-hoc subtype error analysis (Sec. 5.6), never as a training or prediction target.

The prompt-design grid is a curated 18 of 32 possible component combinations (Sec. 3.1), not an exhaustive search, and every component's wording is hand-designed rather than automatically optimized or theoretically grounded; this includes the aspects component's seven linguistic markers, which reflect our own judgment rather than an established sociolinguistic framework. A negative result within this space is evidence about the designs actually tested, not proof that no prompt formulation could close the gap. The few-shot demonstrations for every main-grid variant are also drawn once, with a fixed seed; Sec. 3.5's order, count, and demonstration-seed sweeps test sensitivity to arrangement, count, and example selection (Sec. 5.7, Sec. 5.8), but only for each model's own best variant, not the full grid.

The model roster is six open-weight, instruction-tuned models in a 3.8–9B parameter range (Table 1); no larger open model and no closed, commercial API model is evaluated, so these findings do not necessarily extend to that setting. Every inference-cost figure (Table 9) is specific to this study's own hardware and batching choices; the direction of Sec. 6.2's cost argument is likely to hold generally, but its exact multiplier is not.

This study compares against a classical machine-learning baseline and two fine-tuned neural baselines, but not against human inter-annotator agreement, so it cannot say how close, or how far, its numbers sit from whatever ceiling human disagreement on sexism labeling itself imposes.

Whether llama3.1's chain-of-thought refusal pattern (Sec. 5.3, Table 3) generalizes to the other five models has been checked systematically: every chain-of-thought cell with at least one failure across mistral, phi3, qwen2.5, gemma2, and qwen3 (47 cells) was classified by the same refusal-pattern rule, measuring explicit-refusal prevalence against the full 4,000-example test split rather than each cell's own failure count, since the two can diverge sharply when `fail_ratio` itself is small. Explicit refusal never exceeds 1.05% of the test split for any of the five other models, while every one of llama3.1's own ten other chain-of-thought cells sits far above that, from 4.82% to 48.1%: the refusal pattern is, on this evidence, a property of that model specifically, not a shared failure mode this study's prompt design triggers generally.

Sec. 5.10's AUC comparison and Sec. 5.11's abstention policy cover only the seven non-chain-of-thought variants: the eleven chain-of-thought variants, including qwen3's own best-F1 variant (V8), have no single decision token this method can score. The abstention gain is also not uniform: phi3 buys a large F1 improvement only by abstaining on over a third of the test split, while the other four models gain little at a much smaller coverage cost.

Generation throughout this study is greedy and seeded, which removes sampling randomness but not kernel-, driver-, or library-level nondeterminism across hardware. Each of five models' best variant was rerun on two cluster nodes sharing a GPU architecture (GA100, compute capability 8.0) but different memory configurations (40GB- and 80GB-per-card): every one of 4,000 test examples produced byte-identical raw output and an identical parsed decision on both nodes, for all five models (the llama3.1 V14 reproduction in Sec. 5.3 is a special case of the same check). This is directly tested evidence for bit-identical determinism within one GPU architecture; whether the same holds across different architectures remains untested.

8 Conclusion

This paper evaluated 18 prompt-design variants across six open-weight, instruction-tuned LLMs on the official EDOS sexism-detection benchmark [7], with paired significance testing, subtype-level error analysis, order, few-shot-size, and demonstration-selection sensitivity sweeps, and a head-to-head comparison against QLoRA, an encoder, and a classical TF-IDF baseline. The clearest result in this study cuts against prompting. Most of this study's prompt-design choices produce large, statistically real effects (92 of 101 comparisons significant under McNemar's test, 88 of 101 under the paired F1-diff bootstrap), and phi3's largest single effect (+0.428 F1) is the biggest shift this study puts through significance testing, though not the biggest shift in the study overall: QLoRA fine-tuning moves the same model further still (Sec. 6.1). But those effects operate entirely within a ceiling that a linear classifier over word counts already clears for five of the six models, at a fraction of the computational cost. Only qwen3 beats that classifier on F1, and it does so at 4–16× the inference cost of every other prompted model, while still losing to it on accuracy. Prompt design is a real, measurable lever on this task, but the evidence here does not make it a substitute for adapting a model, however simple, to the task's own labeled data.

Two further findings deserve separate mention. Llama-3.1-8B's interaction with chain-of-thought prompting is not the single-variant curiosity it first appeared to be: every chain-of-thought variant this model was evaluated on shows elevated generation failures, and the cause splits cleanly depending on whether the prompt also asks for aspect-by-aspect analysis, explicit safety refusal when it does not, reasoning-token-budget exhaustion when it does (Sec. 5.3). Neither failure mode was visible in the aggregate accuracy or F1 numbers alone; both surfaced only once the raw generations were read directly. A model's headline score can look ordinary while concealing a specific, systematic reason it is wrong. Separately, prompt-*component* order turned out to matter far less than prior work on few-shot demonstration order would predict, moving only one of six models' scores in either direction (Sec. 5.7). In this study, which components a prompt includes matters far more than the order they appear in.

For anyone building a sexism-detection or similar content-moderation pipeline on an off-the-shelf instruction-tuned LLM, prompt engineering is not wasted effort: it measurably changes what these models can do. But it should be budgeted and evaluated as one option among several rather than assumed to be the strongest one, and any pipeline that treats a model's silence or refusal as an ordinary negative prediction should monitor for that directly instead of trusting an aggregate score to reveal it. The most direct next steps are the ones this paper's limitations already name: testing whether the same pattern holds for closed, commercial models, and extending confidence-based scoring to chain-of-thought variants, the one part of this study's own grid, including qwen3's best-performing variant, that first-token AUC cannot reach at all.

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